



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 2

Introduction to Programming

SESSION / ASSIGNMENT – 3

TASKS :

1. Declare variables for a Flipkart product: productName (as a string), price (float), and rating (double). Assign sample values and print each variable with its data type.



```
#include <stdio.h>

int main(){
    char productname[]="Spiderman BND Poster t-Shirt XL";
    float price=699.99;
    double rating=4.5;

    printf("Product name : %s. Data Type : String",productname);
    printf("\nPrice : %.2f, Data Type : float",price);
    printf("\nRatings : %.1f, Data Type : double ",rating);
    return 0;
}
```

```
Product name : Spiderman BND Poster t-Shirt XL. Data Type : String
Price : 699.99, Data Type : float
Ratings : 4.5, Data Type : double
```

2. Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.

Constraint: The GST rate must not be changeable after its initial assignment.



```
#include <stdio.h>

int main(){
    const float GST=18.0;
    float price=599.68;
    float total= price + (price * GST / 100);

    printf("Dal Makhni 1 serve : %.2f",price);
    printf("\nGST : %.2f%%",GST);
    printf("\nTotal Price : %.2f",total);

    return 0;
}
```

```
Dal Makhni 1 serve : 599.68
GST : 18.00%
Total Price : 707.62
```

3. Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.



```
#include <stdio.h>

int main(){
    char playlist_name[]="Sad Music";
    int songs_no=8;
    float average_songs=3.2;

    printf("\nThe Playlist \"%s\" has \"%d\" Songs with it's Average Duration of the Songs is\n\"%.1f\" per song.\n ",playlist_name ,songs_no ,average_songs);

    return 0;
}
```

```
The Playlist "Sad Music" has "8" Songs with it's Average Duration of the Songs is "3.2" per song.
```

4. Identify and correct the invalid variable names from this list: 1stPlayer, player_age, \$score, total-marks, userName.

Hint: Apply standard variable naming rules for your programming language.



Variable Names	Valid/Invalid	Correct Variable Name
1stPlayer	Invalid	Player1 or first_player
player_age	Valid	-
\$score	Invalid	score
total-marks	Invalid	Total_marks
userName	Valid	-